DATE:1/09/2026

EXPERIMENT 32

Prolog Program to Implement Pattern Matching

Aim

To write and implement a Prolog program for pattern matching using facts, rules, and variables.

Algorithm

1. Define a set of patterns as facts.
2. Define a predicate to compare the given input with the stored patterns.
3. Prolog uses unification to match the input pattern with the facts.
4. If the pattern matches, Prolog returns yes.
5. If no pattern matches, Prolog returns no.
6. Use backtracking to find all possible matches.

Prolog Code

```
% Pattern Matching in Prolog


% Facts

pattern(cat, animal).

pattern(dog, animal).

pattern(parrot, bird).

pattern(eagle, bird).

pattern(rose, flower).

pattern(lotus, flower).
```

```
% Pattern matching rule
match(X, Y) :-
   pattern(X, Y).
```

Input

```
match(cat, animal).
```

Output

```
(15 ms) yes
| ?- match(cat, animal).

yes
| ?- |
```

Result

Thus, the Prolog program for pattern matching was successfully implemented using facts, variables, unification, and backtracking.